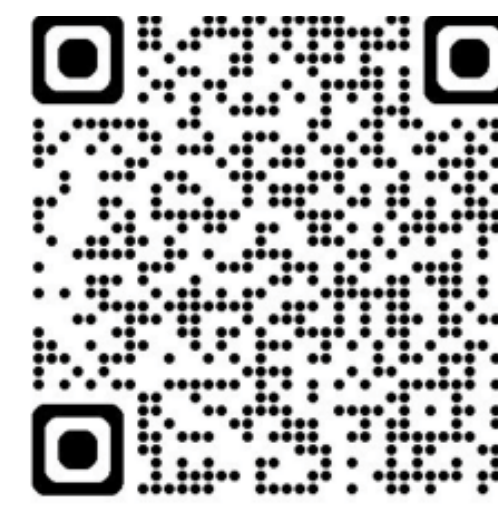




IDAG-Edit: Multi-Object Video Editing via Instance-Decoupled Attention and Guidance

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Background

Motivation

- Precise control in multi-object video scenes
- Disentangled attribute manipulation for each instances
- Seamless identity preservation over temporal dynamics

Problem

- Weak spatial grounding → semantic leakage
- Global guidance → entangled object trajectories
- External temporal constraints → motion misalignment
- First-frame propagation → accumulated errors

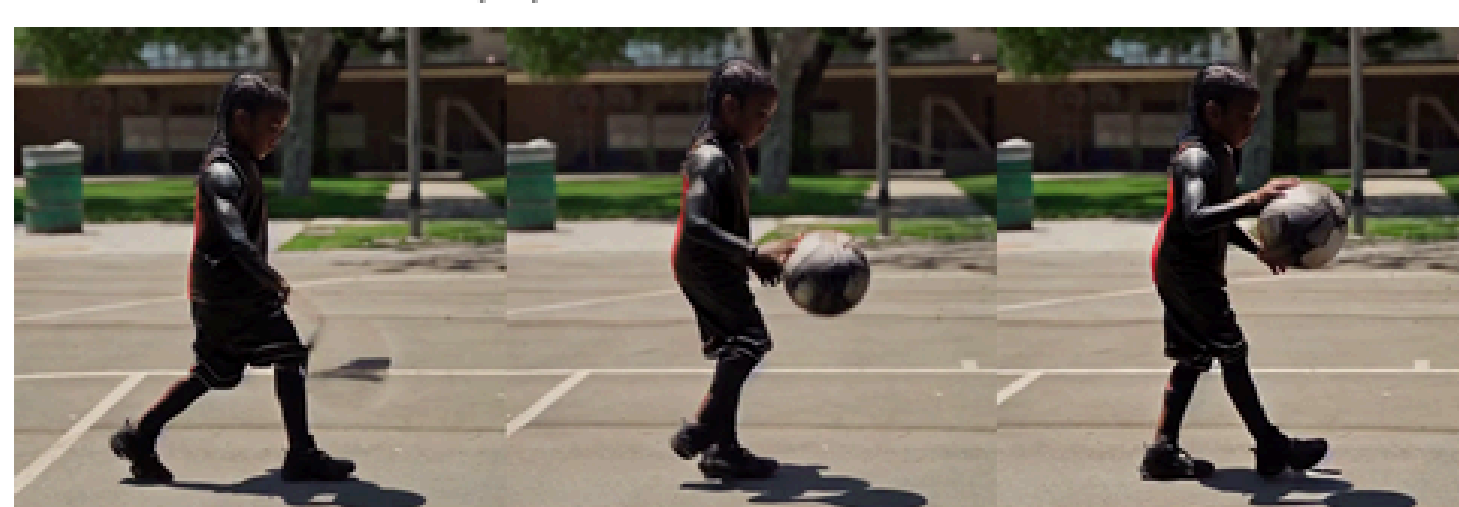
Prompt: "A **batman** in **dark armored suit** dribbles a **soccer** on an outdoor court"



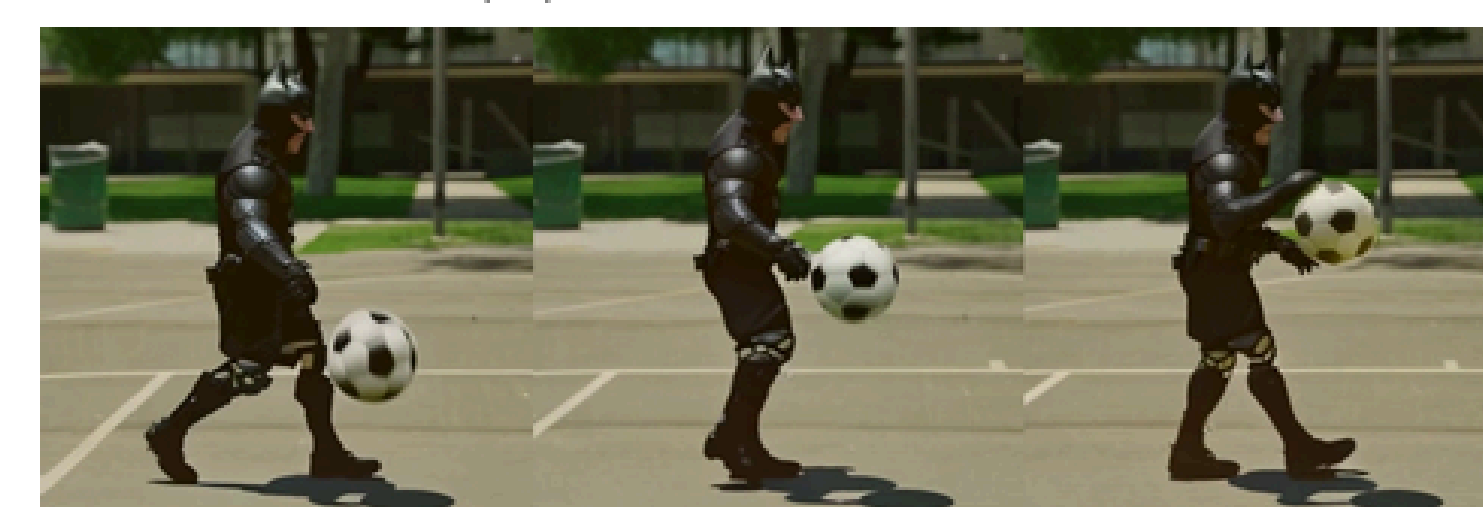
(a) Source video



(b) VideoDirector



(c) VideoGrain



(d) IDAG-Edit (Ours)

Methodology

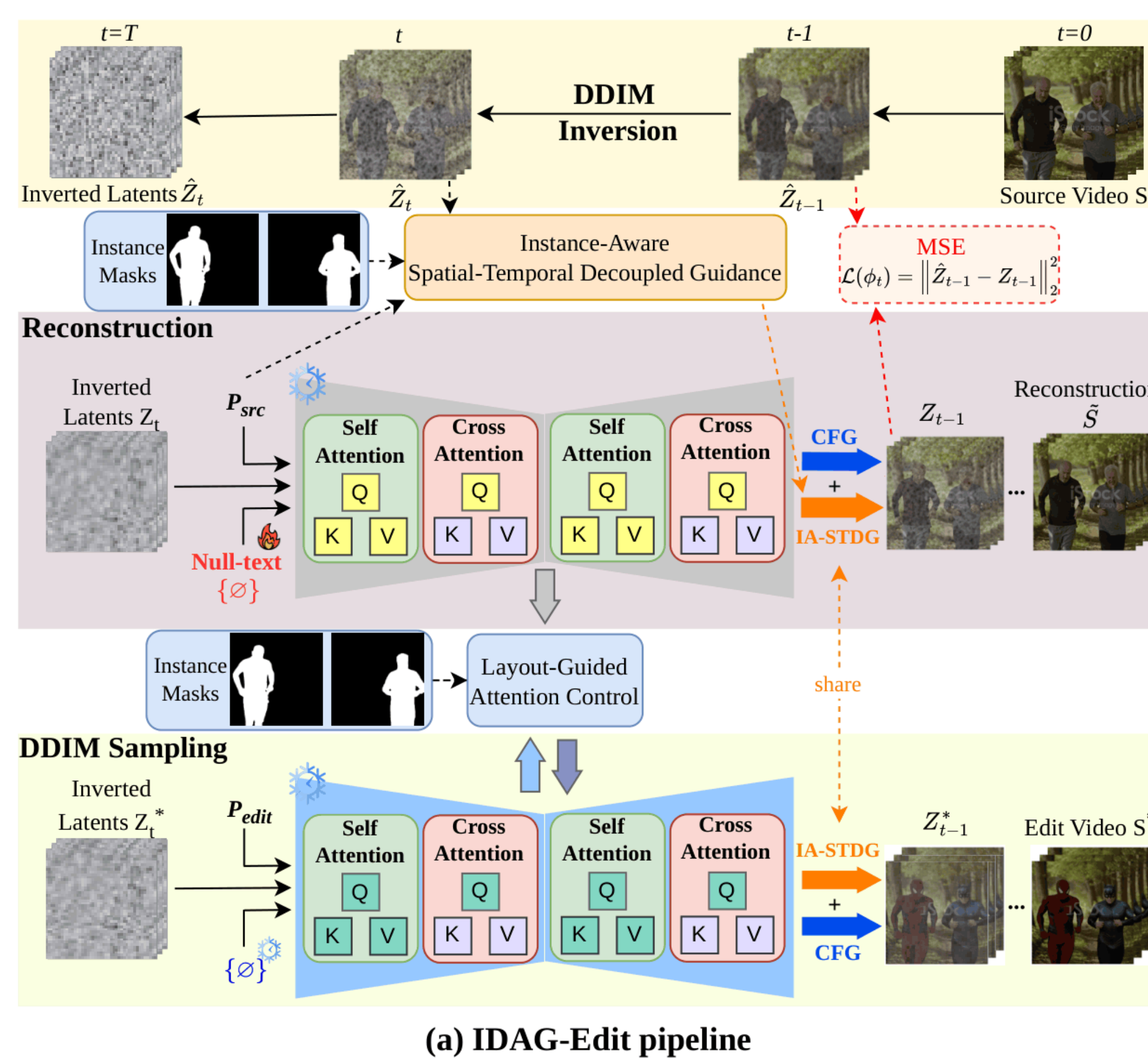
We present IDAG-Edit, enable fine-grained and temporally coherent **multi-object video editing**

IDAG-Edit, a training-free, three-stage editing framework:

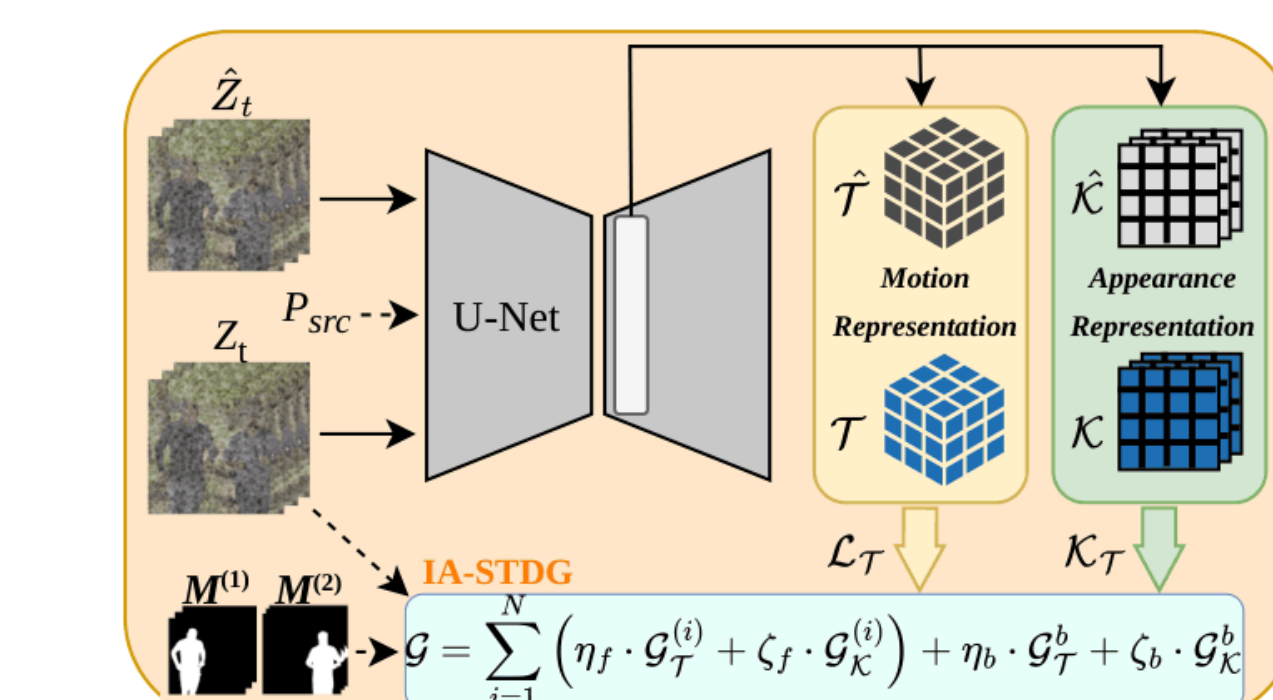
- DDIM Inversion
→ Preserve source structure & motion

- Null-Text Optimization
→ Improve reconstruction fidelity

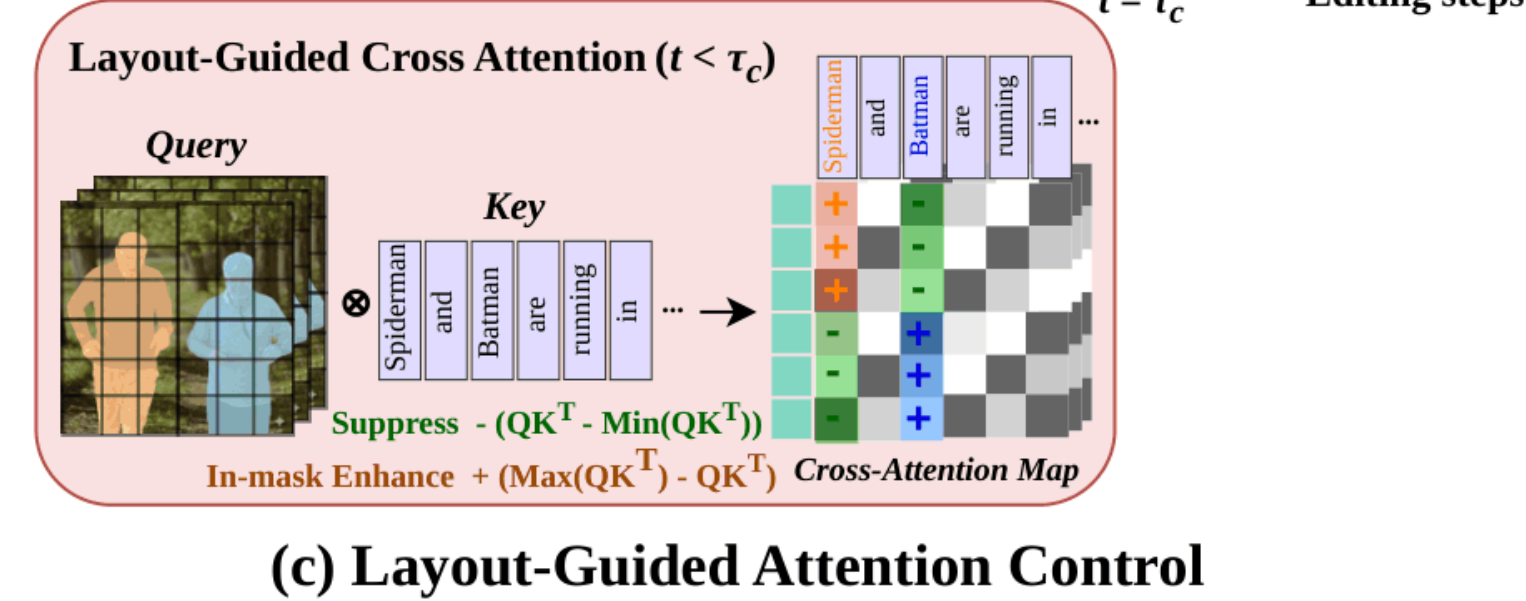
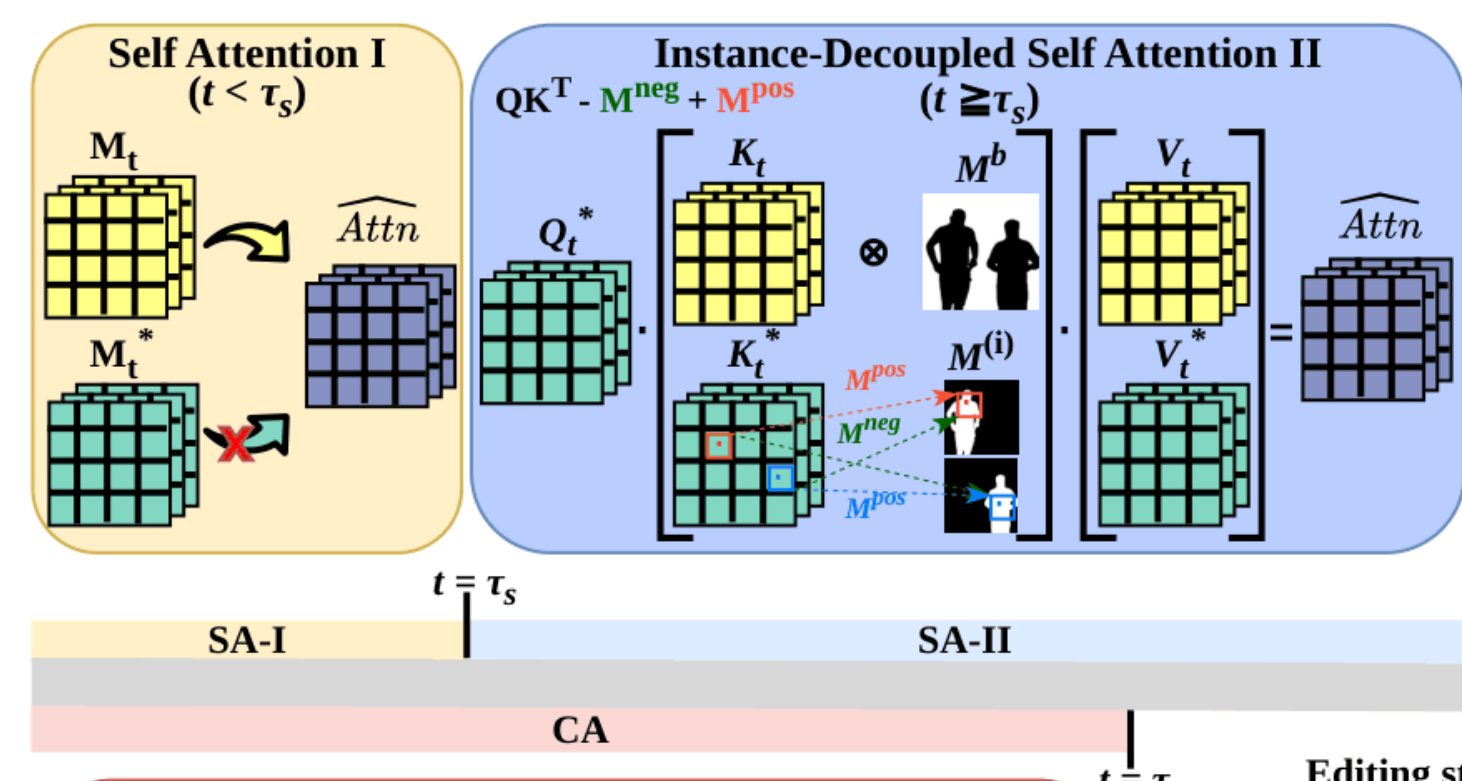
- Instance-Decoupled Editing
→ Prevent cross-instance interference



(a) IDAG-Edit pipeline



(b) Instance-Aware Spatial-Temporal Decoupled Guidance



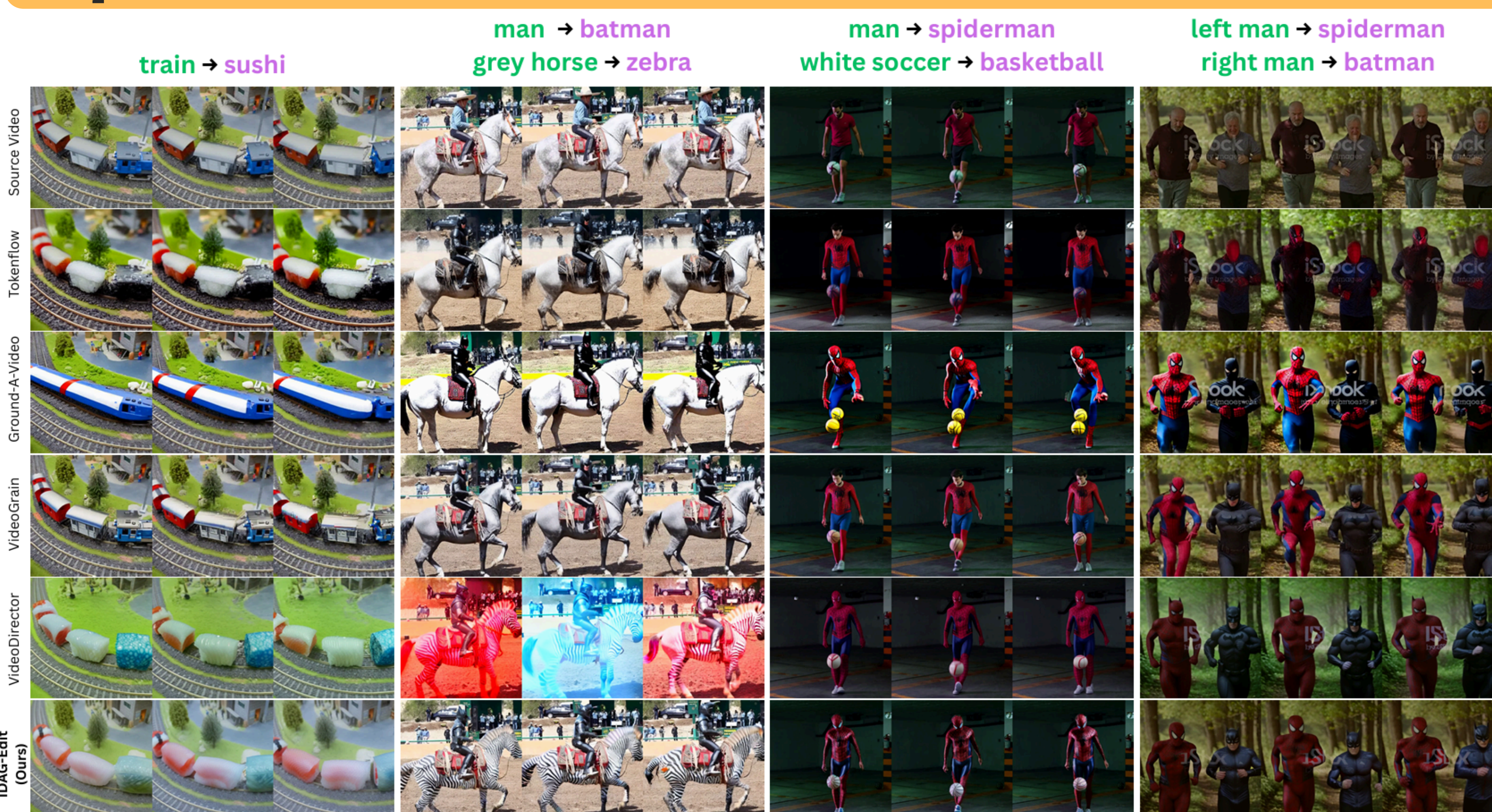
(c) Layout-Guided Attention Control

IA-STDG applies instance-wise spatial-temporal guidance
→ preserves object identity and motion trajectories

ID-SA preserves instance-specific features via mask-constrained self-attention.
→ prevent cross-object feature leakage

LG-CA improves text-region alignment via instance-mask-guided cross-attention.
→ facilitate multi-object control

Experiment Result



- IDAG-Edit improves subject consistency and text alignment, outperforming SOTA in multi-object editing.
- Visual results demonstrate precise object-level editing with minimal semantic leakage, edit failure, or temporal color flickering

Method	Venue	Automatic Metrics (Single-object / Multi-object)					User Study		
		SC ↑	BC ↑	MS ↑	CLIP-F ↑	CLIP-T ↑	Edit-Acc ↑	TC ↑	Overall ↑
TokenFlow	ICLR'24	93.16/93.51	92.23/93.13	96.65/96.32	97.32/97.66	31.32/32.03	2.78	3.28	2.99
Ground-A-Video	ICLR'24	93.02/93.33	92.54/94.03	95.09/95.56	96.72/97.33	32.23/34.43	2.63	2.02	2.25
VideoGrain	ICLR'25	93.69/94.21	94.32/94.81	96.79/96.67	97.43/97.68	31.18/32.46	2.43	2.98	2.58
VideoDirector	CVPR'25	93.39/86.85	93.61/90.73	97.12/94.40	97.40/95.42	31.20/30.33	-	-	-
VideoDirector*	CVPR'25	93.39/93.72	93.61/94.34	97.12/96.91	97.40/97.02	31.20/31.49	2.89	2.83	2.80
IDAG-Edit (Ours)	-	94.13/94.31	94.70/94.83	97.12/97.11	97.44/97.74	31.47/34.76	3.28	3.34	3.23

Conclusion

- Training-free framework for multi-object video editing
- Instance-aware mechanism reduces semantic leakage
- Enables precise object control with better temporal consistency